

1 Surface tension of phase interfaces in ternary mixtures

In this chapter, we investigate the surface tension of interfaces in ternary mixtures composed of three distinct components (labeled 0, 1 and 2). All species have repulsive cross species interactions $\chi_{ij} > 0$. Without loss of generality, we will select component 2 to always have the weakest repulsive interactions with the other two species. As a result component 2 can accumulate at phase interfaces, effectively dampening the interactions between components 0 and 1. We systematically increase the mean concentration of component 2 starting from the dilute limit all the way to the critical point and examine how this dampening affects interfacial properties. We solve the interfacial problem numerically using the computational framework established in the previous section and additionally develop an analytical approximation that captures the essential physics. We then compare the analytical results to the numerical solutions to assess the accuracy and limitations of our theoretical approach. Finally, we derive the critical scaling behavior of the surface tension near the critical point.

1.1 Determining the interfacial profiles and surface tension numerically

1.1.1 Model setup and numerical scheme

After having rescaled the free energy density in Eq. ?? by $k_B T \nu^{-1/3}$ the model only depends on the Flory-Huggins interaction matrix χ_{ij} and the lattice size a . Since the lattice size only sets the interfacial length scale, we fix $a = 1$. With the convention used in Eq. (??), the gradient coefficient matrix is then

$$\kappa_{ij} = -\chi_{ij}. \quad (1)$$

In the following, we restrict attention to interaction parameters:

$$\chi_{12} \in [0.8, 1.5], \quad \chi_{02} = 1.5, \quad \chi_{01} \in [2.3, 3.0] \quad (2)$$

This range reliably produces liquid-liquid phase separation, while avoiding (as far as possible) regions of three-phase coexistence. For a chosen χ -matrix and mean volume fractions, we initialize the system using the corresponding coexisting bulk compositions $\phi_i^{(j)}$ for $i, j \in \{1, 2\}$. These bulk phases are computed using the `flory` package [3] and then used to initialize the interface as a sharp step between the two coexisting phases (see Fig. 1)

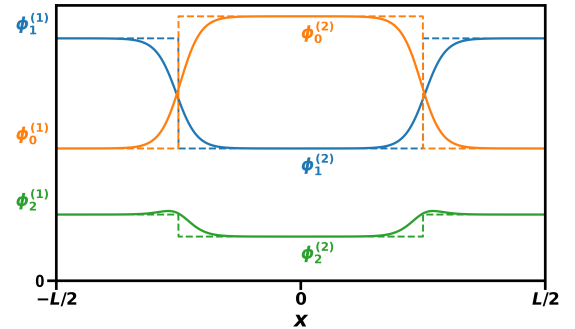


Figure 1: Profile initialization: dashed lines are the initial profiles and solid lines are the equilibrated profiles. Note that we have two interface regions to accommodate the periodic boundaries

The spatial discretization is chosen adaptively to resolve the interface. Thus the grid spacing Δx is set such that the interfacial region spans approximately 40 grid points. The total domain size is 256 grid points such that the fields relax to their bulk values well before the domain edges.

1.1.2 Equilibration dynamics and interfacial structure

Starting from the initial step profile, we relax the system to equilibrium using the fixed-point iteration in Eq. (??). Since we are only interested in the equilibrium profiles, the step size h is chosen as large as possible while maintaining numerical stability. The semi-implicit update strongly suppresses high-wavenumber modes, which allows us to use $h = \mathcal{O}(10^{-1})$, typically one to two orders of magnitude larger than in a fully explicit scheme.

We declare convergence when the maximum change in the volume fractions satisfies

$$\max_x \|\phi^{i+1}(x) - \phi^i(x)\| < 10^{-10}.$$

Depending on interaction parameters and initialization, this criterion is reached after 10^3 - 10^4 iterations.

We continue equilibrating the system for increased mean concentrations of the second species until the critical point beyond which two-phase coexistence no longer is feasible (Fig 2).

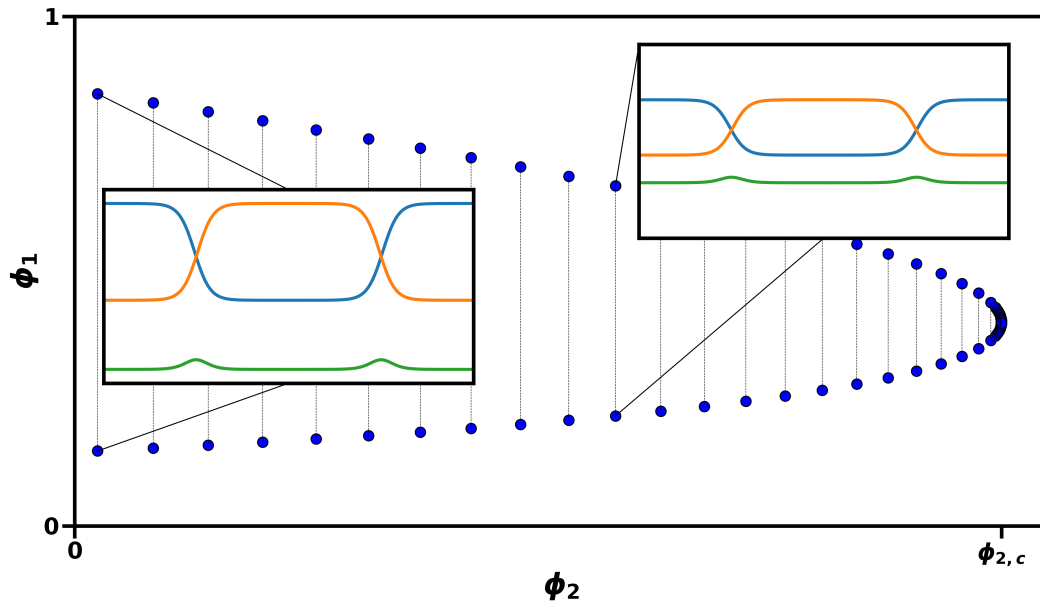


Figure 2: Select mean volume fractions along the binodal in phase space (blue dots) and equilibrate the interfacial profiles for the given mean volume fractions (insets). We start at the dilute limit ($\phi_2 \sim 0$) and repeat this until we reach the critical point ($\phi_2 = \phi_{2,c}$).

At equilibrium, the interface exhibits a smooth, sigmoid-like transition in each component. Species 2 will sometimes be enriched at the interface, thus effectively screening the strongest repulsive interactions. Interestingly the degree of absorption at the interface increases as the repulsive interaction between species 0 and 1 becomes more repulsive and can even lead to a third phase. To give this obser-

vation some merit Huan and Olvera de la Cruz investigated the symmetric case $\chi_{02} = \chi_{12}$, where species 2 is symmetric about the interface. The authors were able to show that the interfacial adsorption scales linearly with the ratio χ_{12}/χ_{01} , which matches our observation.

1.1.3 Critical scaling of the surface tension

Perhaps add a figure in supplement showing numerical fit of simulation data to power law and resulting critical exponent. Once the equilibrium profiles are obtained, the surface tension is computed using the thermodynamic relation derived in the previous section. The resulting surface tensions are shown in Fig. 3.

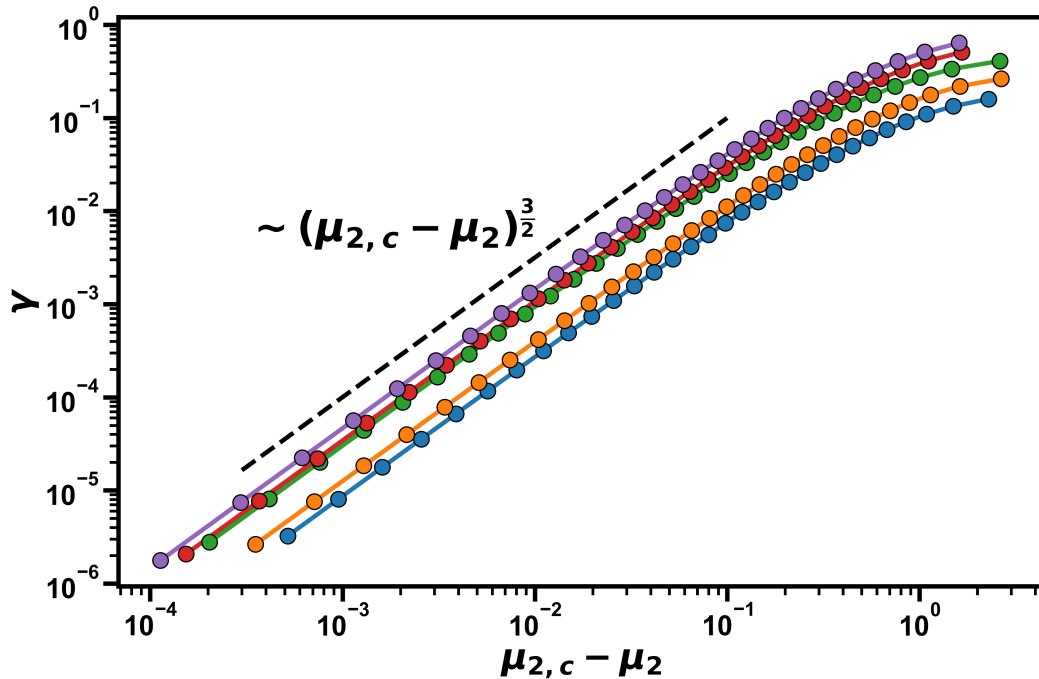


Figure 3: Surface tension as a function of the chemical potential difference $\mu_{2,c} - \mu_2$ for various interaction parameters (colored lines) in the range specified by Eq. (2). The surface tension exhibits a power-law scaling $\gamma \sim (\mu_{2,c} - \mu_2)^{3/2}$ close to criticality (dashed line).

As the system approaches the critical point, the interface continuously vanishes, which explains why the surface tension vanishes as well.

Close to criticality, we observe a clear power-law dependence of the surface tension on the distance from the critical point, which we quantify by the chemical potential difference $\mu_2 - \mu_{2,c}$:

$$\gamma \sim (\mu_2 - \mu_{2,c})^\delta.$$

This scaling is observed for a wide range of interaction parameters and over several orders of magnitude as can be seen in Fig. 3.

Within numerical uncertainty, the critical exponent is universal across all interaction parameters considered and takes a value $\delta \approx 3/2$. This has been confirmed with numerical fits for each individual curve for datapoints close enough to critical-

ity¹. This is consistent with the classical mean-field prediction for binary mixtures derived by Cahn [?]. In contrast, the prefactor of the power law is not universal and varies substantially with the choice of interaction parameters (almost an order of magnitude in Fig. 3).

1.2 An effective interfacial model to approximate the surface tension

While the critical scaling of the surface tension near the critical point provides important physical insight, obtaining a closed-form expression for γ from the full ternary Flory-Huggins free energy is as far as we know intractable. Consequently, for each choice of interaction parameters and mean volume fractions, the surface tension must be computed by solving the full numerical interfacial problem.

To avoid this computational cost, we now introduce an analytical approximation that estimates the surface tension using only the bulk volume fractions of the coexisting phases and the interaction parameters, therefore removing the need to equilibrate the spatially extended system. The approximation remains quantitatively accurate over a broad range of interaction parameters and has the correct scaling near the critical point, as we will demonstrate later.

1.2.1 Derivation of a piecewise-linear interface

Perhaps introduce invariant quantity and then introduce the approximation. For example "To begin we can find a conserved quantity in our spatial system, which we will include in our approximation later."

¹critical exponent typically deviates more than 0.1 from 3/2 if points are included in the fit that lay further than 10^{-2} from the critical point away.

As shown in the previous sections, interfacial profiles in ternary mixtures can be highly non-trivial. To make progress, we approximate the interface by a piecewise-linear profile that connects the two coexisting bulk phases over a finite interfacial width. The interfacial width is set to be the same for all species as can be seen in Fig 4.

This approximation neglects two important features of the exact solution. First, it assumes a common interfacial width for all components, which removes the possibility of adsorption at the interface. Second, it ignores the intrinsic nonlinearity of the equilibrium profiles. We will discuss the implications of these assumptions later on.

We will now derive the closed form expression for the surface tension. To begin, we note that for a one-dimensional interface at equilibrium, the Euler–Lagrange equations imply the existence of a conserved quantity across the interface.

Writing the free energy density as a sum of a local and a gradient contribution and adopting Einstein summation convention, we define

$$f(x) = f_{\text{local}}(\{\phi_i(x)\}) + \frac{1}{2} \phi'_i(x) \kappa_{ij} \phi'_j(x). \quad (3)$$

Introducing the exchange chemical potentials μ_i and osmotic pressure Π , the combination

$$f_{\text{local}}(\{\phi_i(x)\}) - \mu_i \phi_i(x) + \Pi - \frac{1}{2} \phi'_i(x) \kappa_{ij} \phi'_j(x) =: -V(\{\phi_i(x)\}) - T(\{\phi'_i(x)\}) \quad (4)$$

is invariant across the interface. Here, the terms V and T play roles analogous to potential and kinetic energy in classical mechanics.

To see this invariance we can take the spatial derivative of Eq. (4), which yields

$$\begin{aligned} \frac{d}{dx} \left(f_{\text{local}}(\{\phi_i\}) - \mu_i \phi_i + \Pi - \frac{1}{2} \phi'_i(x) \kappa_{ij} \phi'_j(x) \right) &= \left(\frac{\delta f_{\text{local}}}{\delta \phi_i} - \mu_i - \kappa_{ij} \phi''_j(x) \right) \frac{d\phi_i}{dx} \\ &= 0, \end{aligned}$$

where the expression in parentheses vanishes by virtue of the Euler–Lagrange equations. Consequently, we have found a conserved quantity in Eq. (4).

Shifting focus back to our approximation, we assume that each component varies linearly across an interface of width w , connecting the bulk values $\phi_i^{(1)}$ and $\phi_i^{(2)}$ for $i \in \{1, 2\}$. The magnitude of the gradient at the interface $x = 0$ is therefore approximated by

$$|\phi'_i(0)| \simeq \frac{|\Delta\phi_i|}{w},$$

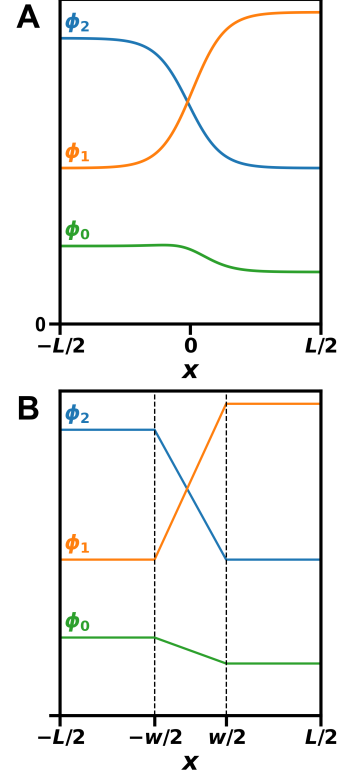


Figure 4: (A) True interfacial profile compared to (B) approximate profile. Perhaps add horizontal lines to indicate the mean volume fractions for each species.

where $\Delta\phi_i = \phi_i^{(2)} - \phi_i^{(1)}$ denotes the difference between the coexisting bulk phases. In our approximation the volume fractions at the interface are equal to the mean compositions

$$\phi_i(0) = \bar{\phi}_i = \frac{1}{2} \left(\phi_i^{(1)} + \phi_i^{(2)} \right),$$

at which the local free energy density is evaluated.

Substituting the gradient approximation into the quadratic gradient term yields

$$\phi_i'(0) \kappa_{ij} \phi_j'(0) = w^{-2} \Delta\phi_i \kappa_{ij} \Delta\phi_j.$$

Solving for the interfacial width gives

$$w = \sqrt{\frac{\Delta\phi_i \kappa_{ij} \Delta\phi_j}{\phi_i'(0) \kappa_{ij} \phi_j'(0)}}.$$

The denominator can be expressed using the invariant quantity Eq. (4), evaluated at $x = 0$. This leads to the approximate expression

$$w = \sqrt{\frac{\Delta\phi_i \kappa_{ij} \Delta\phi_j}{2 (f_{\text{local}}(\{\bar{\phi}_i\}) - \mu_i \bar{\phi}_i + \Pi)}} \quad (5)$$

The surface tension is given by Eq ??

$$\begin{aligned} \gamma &= \int_{-L/2}^{L/2} dx \left[f(\phi_1(x), \phi_2(x)) - \sum_i \mu_i \phi_i(x) + \Pi \right] \\ &= \int dx \phi_i'(x) \kappa_{ij} \phi_j'(x) \end{aligned}$$

where in the second equality we used the invariant quantity from Eq 4.

Within the linear-profile approximation, the gradients are constant across the interface and vanish in the bulk. The integral therefore reduces to

$$\gamma = w \phi_i'(0) \kappa_{ij} \phi_j'(0),$$

Substituting the expression for w from Eq. (5) yields the approximate surface tension

$$\gamma = \sqrt{2 (\Delta\phi_i \kappa_{ij} \Delta\phi_j) (f_{\text{local}}(\{\bar{\phi}_i\}) - \mu_i \bar{\phi}_i + \Pi)} \quad (6)$$

As anticipated, this expression depends only on $\Delta\phi_i$ and $\bar{\phi}_i$ which in turn depend on the bulk compositions $\{\phi_i^{(1)}, \phi_i^{(2)}\}$ of the coexisting phases. Importantly, the closed-form nature of Eq. (6) allows derivatives of the surface tension with respect to order parameters to be evaluated analytically.

The quantitative comparison between the analytical approximation and the numerical results is shown in Fig. 5.

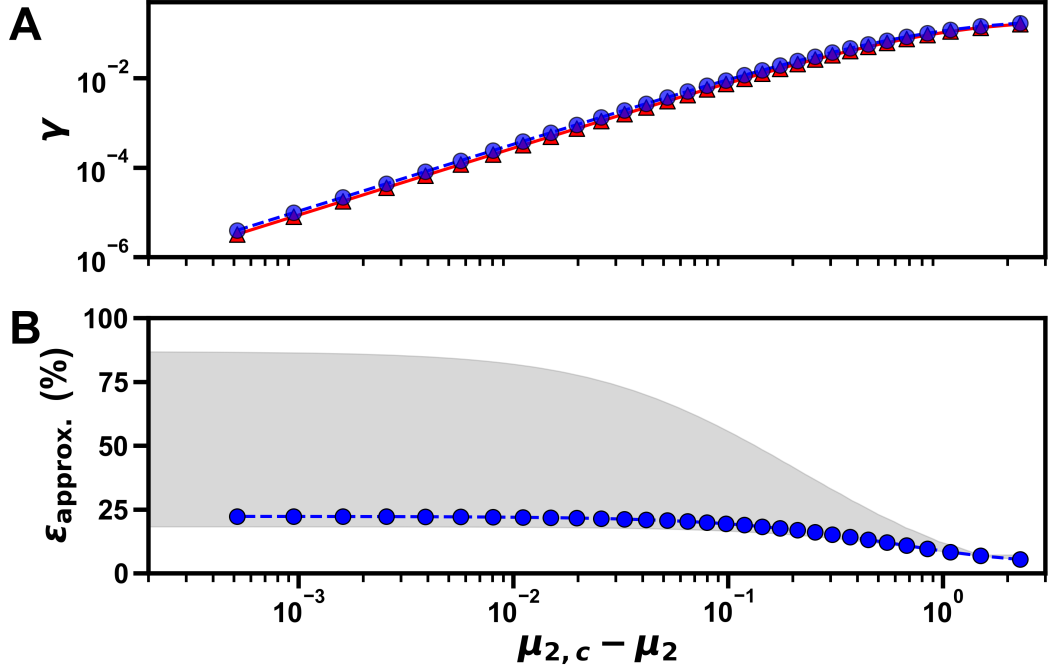


Figure 5: (A) Exact (red) and approximate (blue) surface tension for $\chi_{01} = 2.5$, $\chi_{02} = 1.5$, $\chi_{12} = 1.4$ as a function of the chemical potential. (B) Relative error $\epsilon = (\gamma_{\text{approx}} - \gamma_{\text{exact}})/\gamma_{\text{exact}}$ where the blue data points are for the same parameters as in (A) and the gray area outlines the range of possible errors for various other interaction parameters given by Eq. (2).

Somewhat surprisingly, the approximate expression not only reproduces the correct critical scaling but also provides good qualitative agreement with the numerically computed surface tension over a wide range of parameters. For the parameter range considered in Eq. (2), the surface tension will vary 20% to 90% from the exact value as can be seen in Fig. 5. Additionally, one can see that the approximation becomes more accurate as the system approaches criticality.

A systematic deviation is nevertheless apparent: the analytical approximation consistently overestimates the exact surface tension. This behavior can be understood better within the variational formulation of the interfacial problem. Using the invariant quantity from Eq. (4), the effective Lagrangian density of the system can be identified as

$$L = T - V = 2T = \phi'_i(x) \kappa_{ij} \phi'_j(x),$$

where T and V denote the gradient and local contributions, respectively. The surface tension can therefore be interpreted as the action associated with the interfacial profile,

$$\gamma = \int dx L \equiv S.$$

Since the true equilibrium interface minimizes this action, any approximate interfacial profile necessarily yields a larger value of γ . The systematic overestimation observed in the analytical approximation is thus a direct consequence of the underlying variational principle.

1.2.2 Asymptotic critical scaling

In the previous section, we showed that the surface tension can be expressed approximately in terms of the bulk compositions of the coexisting phases. We now analyze the behavior of this expression in the vicinity of the critical point, with the goal of deriving its asymptotic scaling form.

Recall that the approximate surface tension is given by Eq. (6) as the product of two factors. We first consider the term

$$\Delta\phi_i \kappa_{ij} \Delta\phi_j,$$

which depends solely on the difference between the bulk compositions of the two coexisting phases.

Close to the critical point, the volume fractions in the dense and dilute phases can be written as small deviations from the critical composition $\phi_{i,c}$,

$$\phi_i^{(1)} = \phi_{i,c} + \frac{1}{2}\Delta\phi_i, \quad \phi_i^{(2)} = \phi_{i,c} - \frac{1}{2}\Delta\phi_i,$$

where $\Delta\phi_i$ parameterizes the displacement of the coexisting phases along the tie line in composition space.

In the following we will switch to vector notation, where for example $\boldsymbol{\mu} = (\mu_1 \mu_2)^T$. Expanding the chemical potentials in the two coexisting phases about the critical point yields

$$\boldsymbol{\mu}^{(1/2)} - \boldsymbol{\mu}_c = \mathbf{H} \Delta\boldsymbol{\phi}^{(1/2)} + \mathcal{O}(\Delta\boldsymbol{\phi}^2), \quad (7)$$

where $\mathbf{H}$ denotes the Hessian of the local free energy density evaluated at the critical composition and $\Delta\boldsymbol{\phi}^{(1/2)} = \boldsymbol{\phi}^{(1/2)} - \boldsymbol{\phi}_c$. At coexistence, the chemical potentials in the two phases must be equal, $\boldsymbol{\mu}^{(1)} = \boldsymbol{\mu}^{(2)}$. Subtracting the two expansions and defining

$$\Delta\boldsymbol{\phi} = \Delta\boldsymbol{\phi}^{(1)} - \Delta\boldsymbol{\phi}^{(2)},$$

we obtain, to leading order,

$$\mathbf{H} \Delta\boldsymbol{\phi} = 0. \quad (8)$$

This shows that the difference between the bulk compositions of the coexisting phases lies in the null space of the Hessian at the critical point.

Furthermore, in the immediate vicinity of the critical point, the binodal is approximately linear. Consequently, the deviations $\Delta\boldsymbol{\phi}^{(1)}$ and $\Delta\boldsymbol{\phi}^{(2)}$ are parallel to $\Delta\boldsymbol{\phi}$ and thus also belong to the null space of $\mathbf{H}$. As a result, the linear term in the Taylor expansion of the chemical potentials vanishes along the coexistence direction and higher-order terms control the leading behavior near criticality.

We now turn to the leading nonvanishing contribution in the Taylor expansion of the chemical potential about the critical point. Expanding to second order yields

$$\boldsymbol{\mu}^{(1/2)} - \boldsymbol{\mu}_c = \frac{1}{2} (\alpha^{(1/2)})^2 \left[\frac{\partial^2 \mu_i}{\partial \phi_j \partial \phi_k} \Delta\phi_j \Delta\phi_k \right] \mathbf{e}_i + \mathcal{O}(\Delta\boldsymbol{\phi}^3), \quad (9)$$

where all derivatives are evaluated at the critical composition and we have introduced

$$\alpha^{(1)} = \frac{|\Delta\boldsymbol{\phi}^{(1)}|}{|\Delta\boldsymbol{\phi}|}, \quad \alpha^{(2)} = \frac{|\Delta\boldsymbol{\phi}^{(2)}|}{|\Delta\boldsymbol{\phi}|}.$$

Since again the chemical potentials must be equal in the two coexisting phases, $\mu^{(1)} = \mu^{(2)}$, the prefactors must satisfy $(\alpha^{(1)})^2 = (\alpha^{(2)})^2$. Additionally the two phases must have displacements with opposite sign along the tie line, which implies

$$\Delta\phi^{(1)} = -\Delta\phi^{(2)} = \frac{1}{2} \Delta\phi.$$

The terms inside of the brackets in Eq. (9) is generically nonzero. Consequently, the leading dependence of the chemical potential deviation on the distance from the critical point is

$$|\mu - \mu_c| \sim |\Delta\phi|^2.$$

This relation is consistent with the numerical results shown in Fig. 6A, where the chemical potential difference $|\mu - \mu_c|$ is plotted against $|\Delta\phi|$ on a log-log scale. Finally, since $\Delta\phi$ lies in the null space of the Hessian at the critical point, the leading nontrivial contribution to the dot product $(\mu - \mu_c)^T \Delta\phi$ is equal to the third derivative of the free energy. This quantity vanishes exactly at the critical point, implying that $\mu - \mu_c$ is orthogonal to the tie lines in (ϕ_1, ϕ_2) composition space.

We now turn to the second factor in the approximation for the surface tension,

$$f(\bar{\phi}) - \sum_i \mu_i \bar{\phi}_i + \Pi = f(\bar{\phi}) - f(\phi^{(1)}) + \sum_i \mu_i (\phi_i^{(1)} - \bar{\phi}_i), \quad (10)$$

Where in the second equality we evaluate the pressure at $\phi = \phi^{(1)}$. For convenience we have shortened the notation of the local free energy density $f_{\text{local}} = f$

Close to the critical point, the dense-phase composition can be written as

$$\phi_i^{(1)} = \phi_{i,c} + \frac{1}{2} \Delta\phi_i,$$

while the midpoint composition satisfies

$$\bar{\phi}_i = \frac{1}{2} (\phi_i^{(1)} + \phi_i^{(2)}) = \phi_{i,c},$$

where the last equality follows from the symmetry $\Delta\phi^{(1)} = -\Delta\phi^{(2)}$.

We now expand the local free energy density about the critical composition ϕ_c . Using Einstein summation convention, we obtain

$$\begin{aligned} f(\phi^{(1)}) &= f(\phi_c) + \frac{1}{2} \mu_{i,c} \Delta\phi_i + \frac{1}{2^2 2!} \Delta\phi_i \underbrace{H_{ij} \Delta\phi_j}_{=0} \\ &+ \frac{1}{2^3 3!} \underbrace{\frac{\partial^3 f}{\partial \phi_i \partial \phi_j \partial \phi_k} \Delta\phi_i \Delta\phi_j \Delta\phi_k}_{=0} \\ &+ \frac{1}{2^4 4!} \frac{\partial^4 f}{\partial \phi_i \partial \phi_j \partial \phi_k \partial \phi_l} \Delta\phi_i \Delta\phi_j \Delta\phi_k \Delta\phi_l + \mathcal{O}(\Delta\phi^5). \end{aligned} \quad (11)$$

The quadratic term vanishes because $\Delta\phi$ lies in the null space of the Hessian at the critical point. The cubic term vanishes because the third derivative of the free energy along the coexistence direction is zero at criticality. Consequently, the leading nonzero contribution in Eq. (10) arises from the quartic term, which is generically finite.

Finally, we revisit the higher-order expansion of the chemical potential. As shown earlier, the first-order contribution vanishes and the second-order term is orthogonal to $\Delta\phi$. Additionally we will see that the zeroth term will cancel with Eq 11. Therefore we will include the third-order term as the first nonzero contribution in the second product. The chemical potential can then be written as

$$\frac{1}{2}\mu_i\Delta\phi_i = \frac{1}{2}\mu_{i,c}\Delta\phi_i + \frac{1}{2^4 \cdot 3!} \frac{\delta^4 f}{\delta\phi_i\delta\phi_j\delta\phi_k\delta\phi_l} \Delta\phi_j\Delta\phi_k\Delta\phi_l + \mathcal{O}(\Delta\phi^5). \quad (12)$$

Collecting terms, we find that the second factor of the surface tension approximation is equal to

$$\frac{1}{2^7} \frac{\delta^4 f}{\delta\phi_i\delta\phi_j\delta\phi_k\delta\phi_l} \Delta\phi_i\Delta\phi_j\Delta\phi_k\Delta\phi_l + \mathcal{O}(\Delta\phi^5). \quad (13)$$

and using the scaling relation between the phase composition difference and the chemical potential we find that

$$f(\bar{\phi}) - \sum_i \mu_i \bar{\phi}_i + \Pi \propto |\Delta\phi|^4 \propto |\boldsymbol{\mu} - \boldsymbol{\mu}_c|^2.$$

This scaling is consistent with the numerical results shown in Fig. 6B, where the second factor of the surface tension approximation is again plotted against the chemical potential difference $|\boldsymbol{\mu} - \boldsymbol{\mu}_c|$ on a log-log scale.

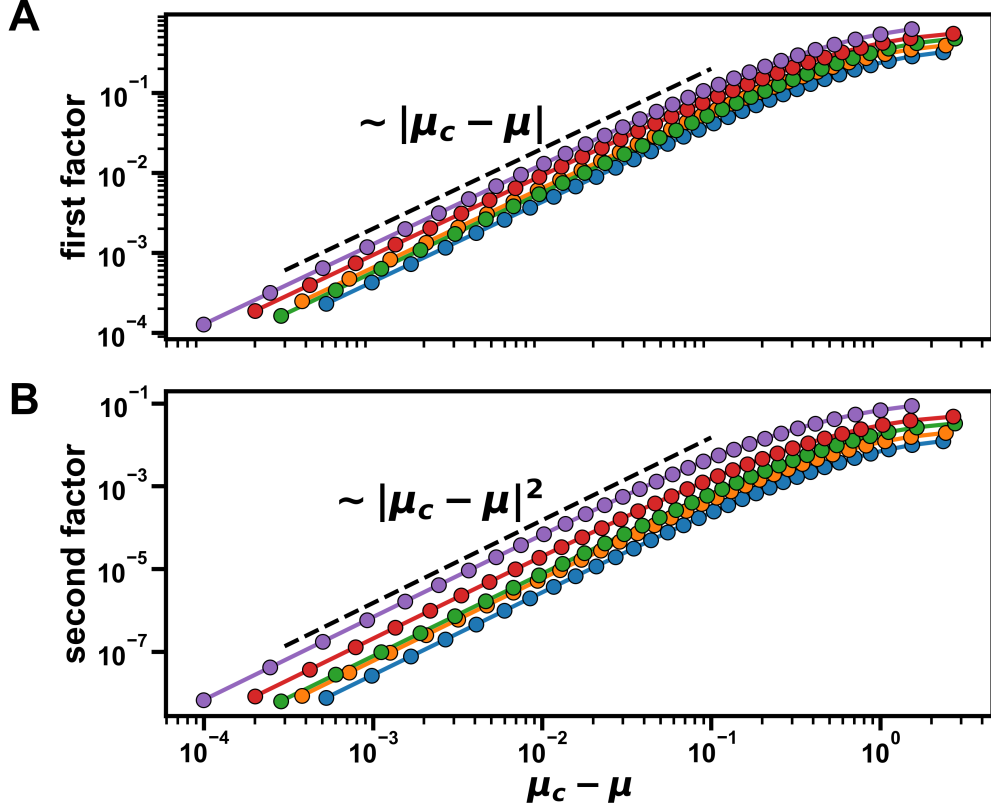


Figure 6: Scaling of the first $|\Delta\phi|^2$ and second $|f(\bar{\phi}) - \sum_i \mu_i \bar{\phi}_i + \Pi|$ term in the surface tension approximation, obtained from the numerical results as a function of the chemical potential difference $\mu_2 - \mu_{2,c}$ for various interaction parameters. The dashed lines indicate the expected scaling with exponents 1 and 2 respectively.

Combining this result with the scaling of the first factor, we conclude that near the critical point the surface tension obeys

$$\gamma \propto |\mu - \mu_c|^{3/2}.$$

In addition, this analysis provides the scaling of the interfacial width:

$$w = \frac{\sqrt{\Delta\phi_i \kappa_{ij} \Delta\phi_j}}{\sqrt{\phi'_i(0) \kappa_{ij} \phi'_j(0)}} \propto |\mu - \mu_c|^{-1/2}.$$

Which diverges at the critical point, as expected.

Taken together, these results provide a direct connection between the bulk thermodynamic quantities introduced in the previous chapter and the interfacial properties analyzed here. The critical scaling of the interfacial width $w \sim |\mu - \mu_c|^{-1/2}$ agrees with the diverging interfacial width observed in the numerical results [Add figure in supplementary](#), while the critical exponent of the surface tension $\gamma \sim |\mu - \mu_c|^{3/2}$ is consistent with our numerically obtained critical exponent. In this sense the effective interfacial model does not introduce new critical behavior, but rather implies that near the critical point the interfacial properties do not depend as much on the profile shape as they do on the bulk compositions.

1.2.3 Sources of error and limitations

The piecewise-linear approximation differs from the true equilibrium interfacial profiles in two essential ways: First, it replaces the smooth, nonlinear variation of the volume fractions across the interface by a linear interpolation. Second, it assumes that all species share a common interfacial width, thereby neglecting interfacial adsorption.

To quantify the importance of species dependent interfacial widths, we fit a hyperbolic tangent profile of the form

$$\phi(x) = \frac{\Delta\phi}{2} \tanh\left(\frac{x - x_0}{w}\right) + \bar{\phi}$$

to the numerical equilibrium profiles of both component 1 and component 0 (Fig. 7A). The difference between the fitted interface positions x_0 for the two species then provides a direct measure of the deviation from the equal-width assumption and thus of the degree of interfacial adsorption.

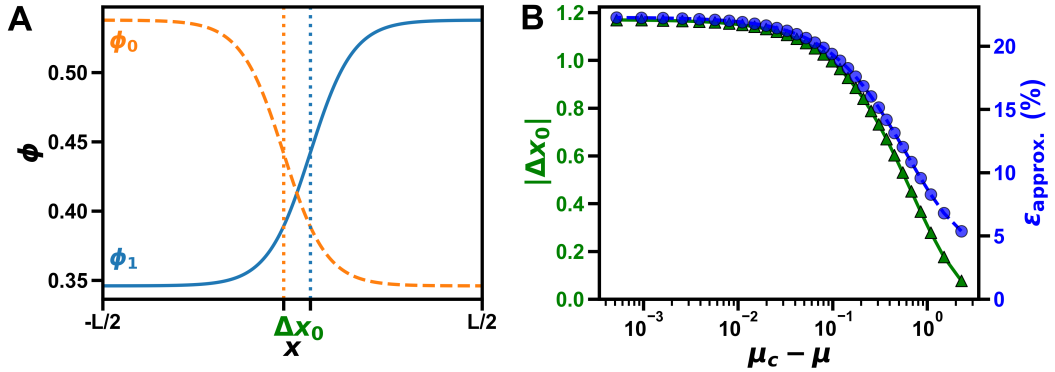


Figure 7: (A) Fitting a hyperbolic tangent profile to the numerical equilibrium profiles to obtain the interface positions x_0 for both components. (B) Relative error of the surface tension approximation (blue) and displacement between the centers of the profiles (green) as a function of the chemical potential.

The results are shown in Fig. 7B, together with the relative error of the surface-tension approximation. A clear correlation is observed between the relative error and the displacement between the centers of the profiles of the first two components. Importantly, this correlation is not restricted to the specific interaction parameters shown in Fig. 7, but persists across a broad range of interaction strengths [Add in supplemental material](#).

One might notice that once the relative displacement between the centers of the profiles nearly vanishes, the relative error does not vanish but rather reaches a finite value. This suggests that even in the absence of interfacial adsorption, the simplified functional form of the profile still contributes to the error of the approximation. However, this relative error (approximately 5% in Fig. 7B) is substantially smaller than the error observed close to the critical point (approximately 22% in Fig. 7B), where the displacement between the centers of the profiles is largest. This indicates that interfacial adsorption, rather than the simplified functional form of the profile, is the primary limitation of the piecewise-linear model.

To further investigate the role of interfacial adsorption, we can try to find an analytical expression for the interfacial absorption and include it in the approximation. The hope is that this will significantly reduce the error of the approximation, especially close to the critical point where the relative error is the largest.

1.3 Analytical expressions for interfacial profiles in symmetric ternary mixtures

Although an analytical solution for the interfacial profile is generally not available, it is possible to derive an implicit expressions for the profile properties, for the symmetric case i.e. $\chi_{01} = \chi_{02}$ and $\phi_0^{(1)} = \phi_0^{(2)}$. In this case, the interaction matrix takes the form ²

$$\begin{pmatrix} 0 & \eta\chi & \eta\chi \\ \eta\chi & 0 & \chi \\ \eta\chi & \chi & 0 \end{pmatrix}$$

Where χ is the interaction strength between the last two components and η modulates the interaction strength of the zeroth component with the other two and typically takes values between 0 and 1. This writing convention has been adopted from other work on interfacial absorption [1, 2] and allows for a direct comparison with their results.

1.3.1 Implicit expression for the bulk phase compositions

The interaction matrix of the reduced two-component system is then given by: **reference equation from theory chapter that shows how reduced matrix is computed from 3 comp system.**

$$\begin{pmatrix} \chi_d & \chi_o \\ \chi_o & \chi_d \end{pmatrix}$$

where χ_d, χ_o are the diagonal and off-diagonal elements, respectively. We choose $\lambda = 1$ such that $\kappa_{ij} = -\chi_{ij}$. The profiles are then given by the variational principle $\delta F / \delta \phi_i = 0$, which for the Flory-Huggins free energy in Eq. ?? leads to the following Euler-Lagrange equations:

$$\begin{aligned} -\chi_d \phi_1''(x) - \chi_o \phi_2''(x) &= \chi_d \phi_1 + \chi_o \phi_2 + \ln(\phi_1) - \ln(1 - \phi_1 - \phi_2) - \mu_1 \\ -\chi_o \phi_1''(x) - \chi_d \phi_2''(x) &= \chi_o \phi_1 + \chi_d \phi_2 + \ln(\phi_2) - \ln(1 - \phi_1 - \phi_2) - \mu_2 \end{aligned}$$

Where $\mu_1 = \mu_2 = \mu = \chi_d \phi_1^{(1)} + \chi_o \phi_2^{(1)} + \ln(\phi_1^{(1)}) - \ln(\phi_0^{(1)})$ is the chemical potential of the coexisting phases. The EL equations can be written in terms of the eigenmodes of the interaction matrix $\phi_{\pm} = \frac{1}{2}(\phi_1 \pm \phi_2)$:

$$\begin{aligned} \frac{2}{\phi_s} \phi_+''(x) &= -\frac{2}{\phi_s} \phi_+ + \ln(\phi_+ + \phi_-) + \ln(\phi_+ - \phi_-) - 2 \ln(1 - 2\phi_+) - 2\mu \\ \frac{2}{\phi_c} \phi_-''(x) &= -\frac{2}{\phi_c} \phi_- + \ln(\phi_+ + \phi_-) - \ln(\phi_+ - \phi_-) \end{aligned}$$

²Note that here species 0 will be absorbed at the interface, which is different from before where species 2 was absorbed. This is done to better match the convention used in the work of Huang and Olvera de la Cruz [1, 2].

Where we define $\chi_o - \chi_d =: 1/\phi_c$ and $\chi_o + \chi_d =: -1/\phi_s$.

The chemical potential can be rewritten in terms of the mean volume fraction $\bar{\phi} = (\phi_1^{(1)} + \phi_2^{(1)})/2$ and the difference $\Delta\phi = (\phi_1^{(1)} - \phi_2^{(1)})/2$

$$\mu = -\bar{\phi}/\phi^* - \Delta\phi/\phi_c + \ln(\bar{\phi} + \Delta\phi) - \ln(1 - 2\bar{\phi})$$

First we will focus on the EL equation for ϕ_- . For this we can look at the equation far away from the interface, in the bulk where the curvature is zero and $\phi_+(x \rightarrow \infty) = \bar{\phi}$ and $\phi_-(x \rightarrow \infty) = \Delta\phi$ are constant. Then we get the implicit equation for $\Delta\phi$:

$$\begin{aligned} -\frac{2}{\phi_c}\Delta\phi + \ln(\bar{\phi} + \Delta\phi) - \ln(\bar{\phi} - \Delta\phi) &= 0 \\ \Rightarrow \Delta\phi &= \bar{\phi} \tanh\left(\frac{\Delta\phi}{\phi_c}\right) \end{aligned} \tag{14}$$

By rewriting this expression as

$$x = a \tanh(x),$$

where $x = \Delta\phi/\phi_c$ and $a = \bar{\phi}/\phi_c$ one can make some statements on the possible outcomes. If $a \leq 1$ or $\bar{\phi} \leq \phi_c$ the only root is $x^* = 0$. Therefore phase separation can only occur for $\bar{\phi} > \phi_c$. The larger a gets, the larger the root x^* becomes. The largest possible value is the dilute limit where $\bar{\phi} = 1/2$. This is where $\Delta\phi = \tanh(\Delta\phi/\phi_c)/2$ is the largest as well.

1.3.2 Implicit expression for the interfacial absorption

Next we will look at the EL equation for ϕ_+ at the interface. Due to the anti-symmetric nature of ϕ_- about the interface we have $\phi_-(x=0) = 0$ and therefore

$$\begin{aligned} \frac{2}{\phi_s}\phi_+''(0) &= -\frac{2}{\phi_s}\phi_+(0) + 2\ln(\phi_+(0)) - 2\ln(1 - 2\phi_+(0)) - 2\mu \\ \Rightarrow \frac{\phi_+''(0)}{\phi_s} &= -\frac{\delta\phi_+}{\phi_s} + \frac{\Delta\phi}{\phi_c} + \ln\left(\frac{\bar{\phi} + \delta\phi_+}{\bar{\phi} + \Delta\phi}\right) + \ln\left(\frac{1 - 2\bar{\phi}}{1 - 2(\bar{\phi} + \delta\phi_+)}\right) \end{aligned}$$

Where we have defined $\delta\phi_+ = \phi_+(0) - \bar{\phi}$. This quantity essentially . Now we can rearrange this to the implicit equation for $\delta\phi_+$:

$$\delta\phi_+ = \frac{1 - 2\bar{\phi}}{2} \left(1 - \frac{1}{1 - 2\bar{\phi} + 2(\bar{\phi} + \Delta\phi)\Pi} \right) \tag{15}$$

Where we have defined

$$\Pi = \exp\left(\frac{\delta\phi_+}{\phi_s}(1 + a_2) - \frac{\Delta\phi}{\phi_c}\right) \tag{16}$$

where $a_2 = \delta\phi_+''(x)/\delta\phi_+$ measures the curvature of ϕ_+ at the interface center. Dimensional analysis then implies

$$\phi_+''(x) \sim \frac{\delta\phi_+}{w^2}$$

where w is the interfacial width. Since the interfacial width diverges at the critical point, we can conclude that in this limit

$$a_2 = \frac{\phi_+''(x)}{\delta\phi_+} \sim \frac{1}{w^2}$$

vanishes. Under the assumption $a_2 \ll 1$ the equation in (15) can be solved self-consistently once $\Delta\phi$ is obtained from (14). The results can be seen in figure ??.

1.3.3 Surface tension approximation with interfacial absorption

Next we can look at the interface width, $w = \Delta\phi / \nabla\phi_{-,IF}$, where $\nabla\phi_{-,IF}$ is the gradient of ϕ_- at the interface center. At the interface we have $\phi_- = \nabla\phi_+ = 0$. We can use the conserved quantity from Eq. 4 to rewrite this as

$$f - 2\mu\phi_+ + \Pi = 2\phi_+ \ln(\phi_+) + (1 - 2\phi_+) \ln(1 - 2\phi_+) - \frac{\phi_+^2}{\phi_s} - 2\mu\phi_+ + \Pi = \frac{(\nabla\phi_-)^2}{\phi_c}$$

$$\phi_-'(0) = \left\{ \phi_c \left[2\phi_{+,0} \ln(\phi_{+,0}) + (1 - 2\phi_{+,0}) \ln(1 - 2\phi_{+,0}) - \frac{\phi_{+,0}^2}{\phi_s} \right. \right. \\ \left. - (\bar{\phi} + \Delta\phi) \ln(\bar{\phi} + \Delta\phi) - (\bar{\phi} - \Delta\phi) \ln(\bar{\phi} - \Delta\phi) \right. \\ \left. - (1 - 2\bar{\phi}) \ln(1 - 2\bar{\phi}) + \frac{\bar{\phi}^2}{\phi_s} + \frac{\Delta\phi^2}{\phi_c} \right. \\ \left. - 2(\phi_{+,0} - \bar{\phi}) \left(-\frac{\bar{\phi}}{\phi_s} - \frac{\Delta\phi}{\phi_c} + \ln(\bar{\phi} + \Delta\phi) - \ln(1 - 2\bar{\phi}) \right) \right] \Bigg\}^{1/2} \quad (17)$$

and thus:

$$w = \sqrt{\frac{\phi_c(\Delta\phi)^2}{2\phi_+ \ln(\phi_+) + (1 - 2\phi_+) \ln(1 - 2\phi_+) - \frac{\phi_+^2}{\phi_s} - 2\mu\phi_+ + \Pi}} \quad (18)$$

The analytical approximation for the interface width can be seen in figure ??. Here we can see that including the absorption yields a much better agreement with the numerical results.

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